

1D and 2D GRF : Using Fourier Space

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Objectives

In this practical session we will explore the relation between a power spectrum in Fourier space and the statistical properties of a random field in real space.

The main goal is to understand that a power spectrum does not define a unique realization of a field. Instead, it specifies statistical properties that are shared by an ensemble of realizations.

By the end of the practical you should be able to:

- generate Gaussian random fields with a prescribed power spectrum;
- understand the role of amplitudes and phases in Fourier space;
- identify the condition required for a Fourier-space field to correspond to a real field in real space;
- understand how the shape of $P(k)$ affects spatial structure;
- connect large and small spatial scales with low and high Fourier modes;
- apply simple Fourier-space filters;
- extend the construction of Gaussian random fields from one to two dimensions;
- understand the meaning of statistical isotropy in two dimensions.

General instructions

The objective of this practical is not only to produce the requested plots, but to understand why they have the behavior you observe.

For several parts of the practical, you will be asked to make a prediction **before running your code**. Record your prediction before performing the numerical calculation.

You are allowed to use AI tools; however, you are responsible for understanding every part of the work you submit. During the practical session, the instructors may ask you to explain, justify, or modify any part of your calculation.

When using random numbers, store and report the random seed used for your main results so that your calculation can be reproduced.

1 One power spectrum, many realizations

Consider a one-dimensional real-space interval

$$0 \leq x \leq 10 \tag{1}$$

sampled using $N = 10000$ points.

Construct the corresponding Fourier frequencies using

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np.fft.fftfreq()
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with the correct real-space sampling interval.
Consider the power spectrum

$$P(k) = (|k| + \epsilon)^{-2}, \quad (2)$$

with

$$\epsilon = 0.1. \quad (3)$$

Before generating the fields

Answer the following questions before writing the random-field generator:

1. If five fields are generated from exactly the same power spectrum, should the five real-space fields be identical?
2. Which properties do you expect the five realizations to share?
3. Which properties can differ between realizations?
4. In Fourier space, what information is associated with the amplitude of a mode and what information is associated with its phase?

1.1 Generate five realizations

Generate five independent random realizations of the Fourier coefficients

$$\delta_k, \quad (4)$$

consistent with the prescribed $P(k)$ and with a real-valued field $\delta(x)$.

Transform each realization back to real space using an inverse FFT and plot the five fields.

Discuss:

1. Are the realizations different?
2. Do they appear to come from the same statistical process?
3. Which visual properties seem similar among the realizations?

2 What condition makes the field real?

A real-space field must satisfy a particular symmetry in Fourier space.

Instead of imposing this condition immediately, first investigate what happens when it is not satisfied.

Generate random Fourier coefficients independently for positive and negative frequencies, without imposing any relation between them.

Perform the inverse Fourier transform.

Questions

1. Is the resulting field real?
2. Inspect its real and imaginary components.
3. Determine the relation that the Fourier coefficients must satisfy in order for $\delta(x)$ to be real.

Write this condition mathematically.

Then modify your construction so that the condition is satisfied and verify numerically that the imaginary component of the inverse transform is negligible.

Conceptual question

Is it sufficient for every Fourier coefficient δ_k to be purely real in order to guarantee that $\delta(x)$ is real?

Explain your answer.

3 Same random phases, different power spectra

We now study how the shape of the power spectrum affects the appearance of the field.

Use a one-dimensional interval

$$0 \leq x \leq 10 \quad (5)$$

sampled using $N = 1000$ points.

Consider

$$P_2(k) = (|k| + \epsilon)^{-2}, \quad (6)$$

$$P_3(k) = (|k| + \epsilon)^{-3}, \quad (7)$$

$$P_4(k) = (|k| + \epsilon)^{-4}, \quad (8)$$

with

$$\epsilon = 0.001. \quad (9)$$

Use the **same underlying random realization** for the three cases. In particular, keep the random phases fixed so that the systematic difference between the three real-space fields is produced by the different power spectra.

Prediction

Before generating the fields:

1. Which of the three fields do you expect to appear smoother?
2. Which one should contain relatively more small-scale structure?
3. Explain your prediction by considering how much power each spectrum assigns to large and small values of $|k|$.

3.1 Numerical comparison

Generate the three fields and plot them together or in separate panels.

Compare their morphology.

In addition, compute a simple quantitative measure of small-scale structure, for example

$$\left\langle \left(\frac{d\delta}{dx} \right)^2 \right\rangle. \quad (10)$$

Compare this quantity for the three fields.

Discuss whether your numerical results agree with your initial prediction.

4 Fourier filtering

Take one of your one-dimensional realizations and introduce a Fourier-space cut at some value k_{cut} .

First remove all modes satisfying

$$|k| > k_{\text{cut}}. \quad (11)$$

Transform the filtered field back to real space.

Before plotting

Predict what will happen to the real-space field.

Which spatial structures should disappear?

Now perform the calculation and compare the filtered and original fields.

4.1 Remove large-scale modes

Repeat the experiment, but now remove the modes satisfying

$$|k| < k_{\text{cut}}. \quad (12)$$

Before transforming back to real space, predict the result.

Compare again with the original field.

Answer:

1. Which Fourier modes are associated with large-scale spatial structure?
2. Which Fourier modes are associated with small-scale structure?
3. Explain the difference between the two filtering operations.

5 Gaussian random fields in two dimensions

We now extend the construction to two dimensions.

Create a two-dimensional real-space domain covering

$$0 \leq x \leq 10, \quad 0 \leq y \leq 10, \quad (13)$$

using a 100×100 grid.

Construct the Fourier frequencies k_x and k_y , and define

$$k = \sqrt{k_x^2 + k_y^2}. \quad (14)$$

Consider the isotropic power spectrum

$$P(k) = (k + \epsilon)^{-2}, \quad (15)$$

with

$$\epsilon = 0.1. \quad (16)$$

Before generating the field

Answer:

1. Why does the power spectrum depend only on $k = \sqrt{k_x^2 + k_y^2}$ rather than separately on k_x and k_y ?
2. What physical or statistical assumption is represented by this choice?
3. If the field is statistically isotropic, should any direction in the (x, y) plane be preferred?

5.1 Generate five 2D realizations

Generate five independent realizations of the two-dimensional Gaussian random field.

Ensure that the Fourier coefficients satisfy the appropriate condition for the real-space field to be real.

Perform the two-dimensional inverse FFT and display the five fields.

Discuss which properties differ between realizations and which properties are shared statistically.

6 Breaking isotropy intentionally

Now replace the isotropic spectrum by an anisotropic model, for example

$$P(k_x, k_y) = \frac{1}{k_x^2 + 4k_y^2 + \epsilon}. \quad (17)$$

Use the same random seed or the same random phases as in one of the isotropic realizations whenever possible.

Prediction

Before generating the field:

1. Is this power spectrum isotropic?
2. Which Fourier-space direction is treated differently?
3. Do you expect the corresponding real-space field to show a preferred direction?
4. Sketch or describe qualitatively what you expect to see.

Generate the field and compare it with the isotropic case.

Discuss whether the real-space morphology agrees with your prediction.

7 Physical interpretation

Answer the following conceptual questions:

1. If two Gaussian random fields have exactly the same theoretical power spectrum, are they necessarily the same realization?
2. What information does the power spectrum specify?
3. What role do Fourier phases play in determining the appearance of a particular realization?
4. Why is the power spectrum particularly important for Gaussian random fields?
5. What information could be missing if the field were not Gaussian?

Good practice: Numerical and consistency checks

It is a good practice to design checks of your calculations. Checks should test a mathematical or physical property rather than simply repeating the calculation with another numerical function. Possible examples include checking:

- the Hermitian symmetry of the Fourier coefficients;
- the size of the imaginary component after the inverse transform;
- whether the recovered spectrum approximately follows the desired shape;
- whether changing the random seed changes the realization but not the intended statistical properties;
- whether filtering Fourier modes produces the expected change in real space.

For each check, you can ask yourself: 1) what you are testing; 2) what result you expect; 3) what result would indicate that something is wrong.

AI-use statement

At the end of your work, briefly state:

- whether you used an AI tool during the practical;
- what you used it for;
- one suggestion or result obtained from the AI that you independently checked;
- how you verified it.

What should you submit?

Submit your notebook or code together with a short report containing:

- your predictions made before the numerical calculations;
- the requested plots and numerical results;
- your interpretation of the effect of changing $P(k)$;
- your explanation of the reality condition;
- your Fourier-filtering results;
- your comparison of isotropic and anisotropic fields;
- your numerical consistency checks;
- your AI-use statement.

The quality of the practical will be evaluated not only from the final plots, but also from your ability to explain why the fields have the statistical and spatial properties you observe.